

Unnamed_Unit

Unit 4 - Code Quest

PDF Documents

PDF Document 1: 1738066626150-Chapter 4_unit 4 - Code Quest.pdf

This PDF document is included in the unit materials.



UNIT 4 CODE QUEST

Applying Knowledge (code & debugging)

CODE

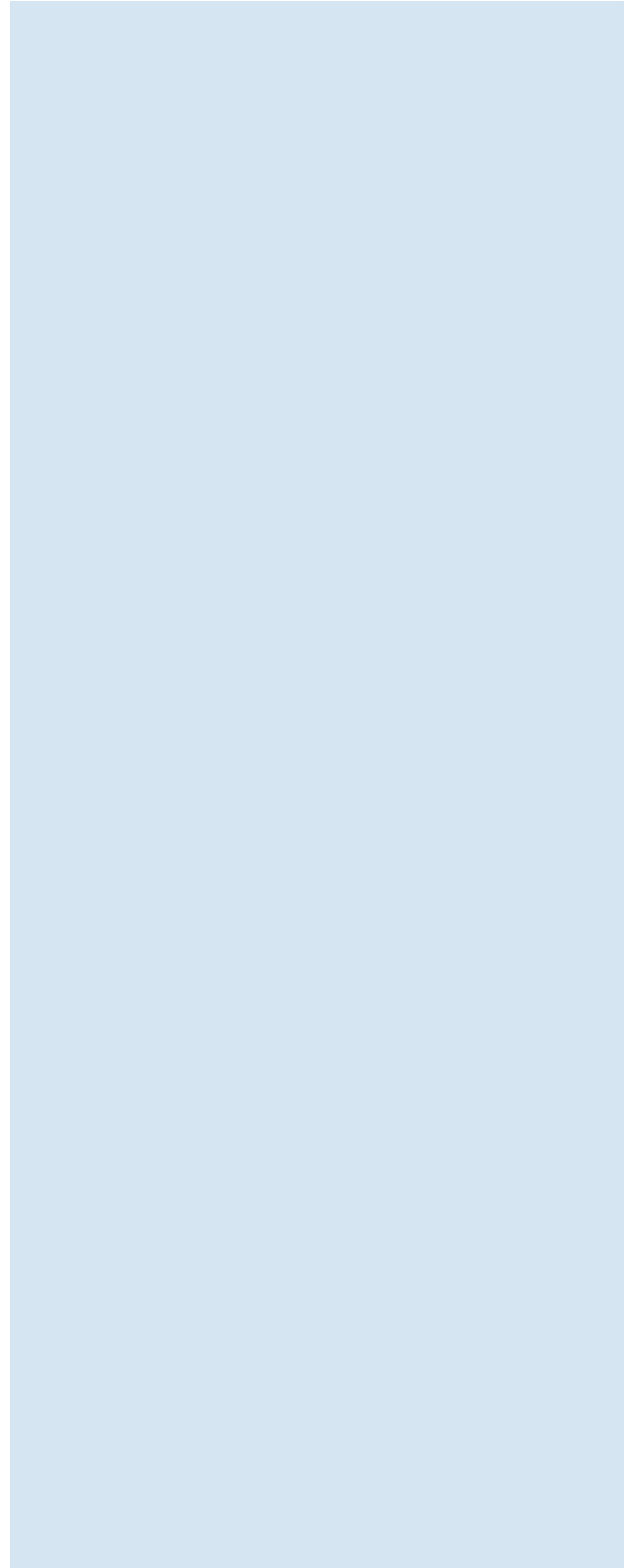
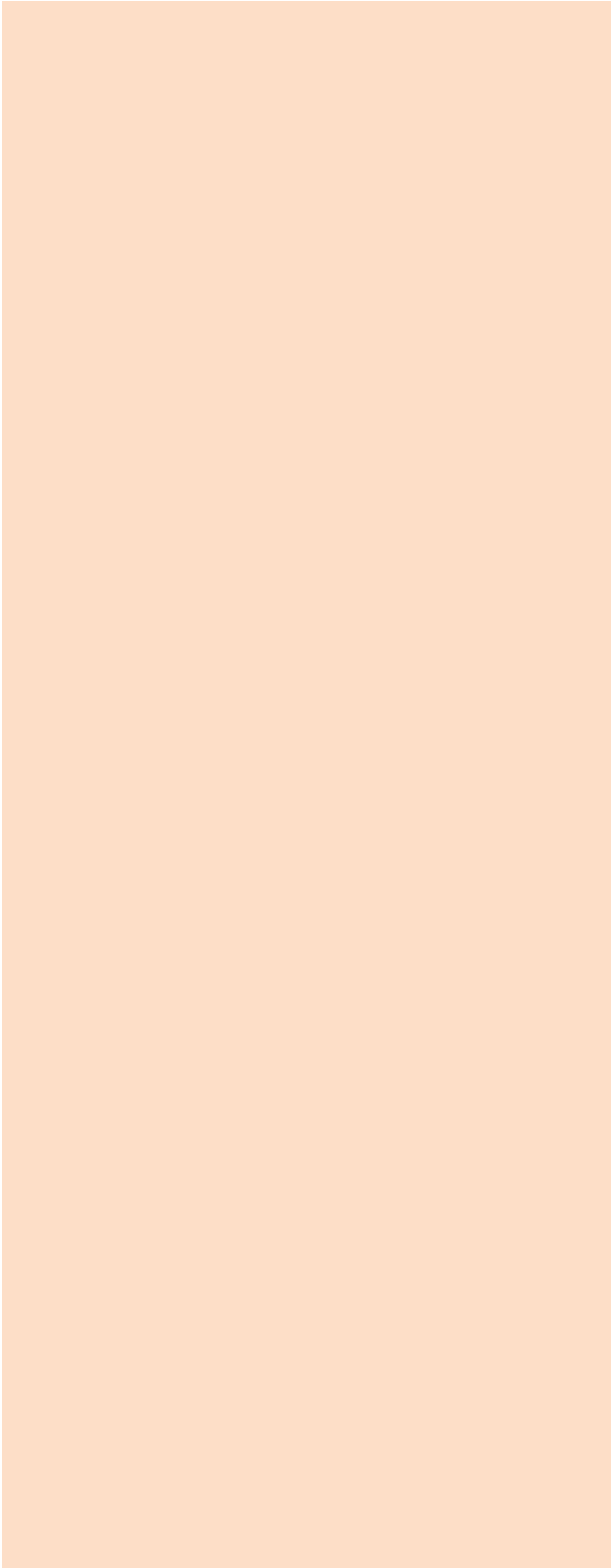




DEBUGGING

There are two "Speaker Off" blocks in the sequence: one at the start and another in the middle of the song.

- 1 What if the "Speaker Off" block at the start is missing?
- 2 What if the "Speaker Off" block in the middle of the song is missing?



Missions

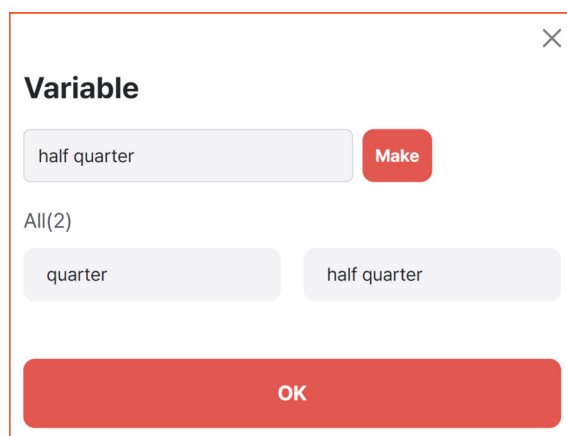
Mission1

1 Let's replace repeating numbers with variables.

There are two different types of wait blocks in the code: ones with a 0.6-second wait and the other with a 1.2-second wait. Let's replace these two numbers with variables.

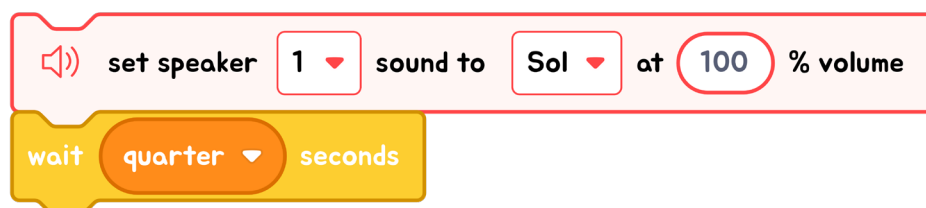
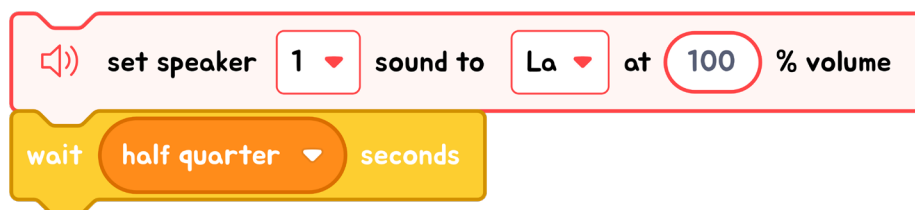
Step1

Create two variables: one for the quarter note and another for the half note. Then, assign the appropriate duration to each variable.



Step2

Replace the value "0.6" in the wait blocks with the variable "quarter" and the value "1.2" with the variable "half."





2 Based on the blocks you've just created, write the corresponding code.

TIPS

It's not that difficult. Once you have created the variables, simply replace the original values in the wait blocks with these variables.



Misson2

- 1 **Let's Finish the Song:** The music sheet contains a lot of repetition. To streamline our code, let's make use of loops to replicate these repeated sections efficiently.

Twinkle, Twinkle, Little Star

DEF#GAB

Traditional

Andante

Twin - kle, twin - kle lit - tle star, How I won - der what you are.

Up a - bove the world so high, Like a dia - mond in the sky!

Twin - kle, twin - kle lit - tle star, How I won - der what you are.

- 2 **Change the Code:** If the code becomes too long and confusing, simplify it by using the comment function. When you press the 'Add Comment' button, a question mark will appear next to the block, where you can enter a comment to clarify what the block does.

1st line

when 1 is clicked then

? set speaker 1 sound to Do at 100 % volume

wait 0.6 seconds